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POLICY AT A GLANCE | 04

Digital Health in India: Unveiling India's eHealth Transformation

Health Sector in India

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Policy at a glance

Digital Health in India : Unveiling India's eHealth Transformation

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The '**Policy-at-a-Glance**' provides users with curated information for first-level analysis and inter-state comparisons of policy. The document brings together all topic-relevant national and state-level acts, policies, and schemes in one place. Further, it identifies for a reader what policy features are covered in which document in a simple table.

The first section of this document sets out some '*Context and Background*' on the topic; the subsequent sections explain the '*Policy Features*' that have been compared; and the last section is the '*Table of Insights*' that marks out what '*Policy Feature*' has been covered (and on which page) in each of the topic-relevant documents.

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1. Context

“Digital Health” represents a wide array of applications. Any use of information technology and communications to manage health and wellness can be considered digital health. The World Health Organisation defines digital health as “a broad umbrella term encompassing eHealth, as well as emerging areas, such as the use of advanced computing sciences in big data, genomics, and artificial intelligence.” (Nishith Desai Associates, 2023)

Digital health technologies rely on health informatics principles to capture, analyse, and utilise health data effectively. By integrating digital health with health informatics, healthcare systems can achieve improved data interoperability, seamless information exchange, and better healthcare decision-making. (National Health Systems Resource Centre, n.d)

Embracing the rapid advancements in technology has bolstered India’s transformative shift in the healthcare sector, along with the potential of revolutionising the sector by harnessing the power of digitalisation and making it more accessible, efficient, and cost-effective. A well-established digital health ecosystem can yield substantial benefits for society, ultimately leading to universal health coverage in India in an efficient, inclusive, affordable, timely, and safe manner. (National Digital Health Footprint, 2019)

Artificial Intelligence (AI), can help overcome the challenges of accessing healthcare facilities, especially in rural areas where there is poor connectivity and a shortage of healthcare professionals. AI can help with early screening, diagnostics, personalised treatment, etc. AI offers great opportunities for large-scale interventions in cancer screening and treatment. India has more than 1 million new cancer cases every year, a number that may rise due to the ageing population and lifestyle changes. Early diagnosis and management can make a big difference in cancer treatment across the country. (National Strategy for Artificial Intelligence, 2018)

Integrating Geographic Information Systems (GIS) can enable spatial tracking of relevant institutions under health care systems. Health planners can use spatial statistics and GIS to evaluate and locate the spatial distribution of diseases among people and make better health planning decisions based on evidence. In past we have successfully used hotspot analysis to identify the highest concentration of HIV/AIDS cases in Churachandpur, Ukhrul, and, Thoubal blocks of Manipur. (Anand et al., 2021)

The Government of India has launched several initiatives for the implementation of digital health in the healthcare process and procedures like Ayushman Bharat Digital Mission, National Telemedicine Services (eSanjeevani and eSanjeevani OPD), etc intending to build a robust “Digital Health Technology Ecosystem.” (National Digital Health Footprint, 2019)

Ayushman Bharat Digital Mission creates a digital health ecosystem that enables smooth and continuous care across different levels of healthcare, from primary to secondary to tertiary. In the digital healthcare ecosystem created by Ayushman Bharat Digital Mission (ABDM), a user has the option to generate a unique Ayushman Bharat Health Account (ABHA) number that has 14 digits. The user can access and manage their health records that are digitally linked to their ABHA number using any health application compatible with ABDM.

To achieve the goals of National Health Policy 2017 and the health-related SDGs, it is important to have a coordinated and integrated approach towards digital health. Therefore, the Government of India has started to create a nationwide and comprehensive digital health ecosystem based on an architectural framework called the National Digital Health Blueprint (NDHB). By implementing NDHB successfully, all Indians would gain a lot of benefits, such as reducing their out-of-pocket expenses, achieving universal health coverage, etc. (National Digital Health Blueprint, 2018)

This document collates information on different policy approaches towards digital health in India. As information technology advances, the transformative shift in the health sector is pertinent for better healthcare decision-making,

and state governments play a pivotal role in shaping policies and digitizing the health sector. Understanding the diverse approaches taken by different states is essential for further development of the sector.

2. Policy Features

The implementation of digital health enables in building of a cohesive health information architecture, health information exchange, and national health information network. When fully implemented, this healthcare system becomes a fully secure and interoperable platform, facilitating the accessibility and portability of health records across both public and private healthcare institutions. Various activities are in progress at the National and the State/UT level for the implementation of digital health across the country. Consequently, there are variations in the extent of coverage across different regions of the country. This section identifies a common set of policy features that the document referred to in section 3 (table of insights) cover. The features are classified under the following three broad aspects of policy: 1) Healthcare Information Storage, 2) Integrated Healthcare Information Management Information System, and 3) Technology-based Digital Supply Chain Management.

1) Healthcare Information Storage

- a. **Maintenance of Electronic Health Records (EHR):** EHRs are digital representations of a patient's medical records, and helps address issues tied to physical records such as loss and lack of accessibility. Ayushman Bharat Digital Mission is a central initiative working towards EHR, but even individual states are making efforts towards digitising the maintenance of records of the patients in hospitals to keep a track of entire patient history, enabling doctors to provide better treatment to their patients. States like Andhra Pradesh, Gujarat, Himachal Pradesh, and others are working towards digitalizing health records by maintaining real-time EHR data. (National Digital Health Mission, 2020)

2) Integrated Healthcare Management Information System

- b. **Automation of Hospital Processes through Health Management Information System (HMIS):** Health Management Information System brings streamlined processes in healthcare facilities, leading to improved healthcare quality, judicious resource utilisation, increased workforce efficiency, and support for research and knowledge sharing, ultimately enhancing the patient experience. HMIS is deployed at a central level and at the state-level state governments have implemented Hospital Information System in the government hospitals. Some of the states that have deployed HMIS program in their government hospitals are Delhi, Goa, and Karnataka among others. (National Health Mission, 2023)
- c. **Integration of Human Resource Management Information System (HRMIS) with Health sector:** The integration of HRMIS involves the incorporation of digital HR tools into healthcare organisations. It streamlines workforce management, enhances personnel data management, and facilitates better resource allocation contributing to an overall improved healthcare system and staff well-being. Some states that have integrated HRMIS within their health sector are Chhattisgarh, Odisha, Madhya Pradesh, and Punjab among others. (Human Resources for Health in District Public Health System of India, 2022)

3) Technology-based Digital Supply Chain Management

- d. **Integration of Information system for eProcurement of medicines:** eProcurement of medicines means to form a systematic approach to procure, manage inventory, and implement Management Information System. The

integration professionalises medical store organisation by leveraging technology for efficient procurement, inventory management, and the overall strengthening of healthcare service. States like Gujarat, Kerala, Maharashtra, Rajasthan, Tamil Nadu, and West Bengal are handling the eProcurement of medicines in their states through information technology. (eAudshadhi, n.d)

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3. Table of Insights: Policy Initiatives for Digital Health

Note 1: Documents linked in the tables include original Acts, Policies, Schemes, Programmes and Guidelines. It is aimed at highlighting and enabling a comparison of relevant policy features covered in the documents, across the centre and the states. While all important documents have been identified, the coverage may not be exhaustive. Further, the 'Policy-at-a-Glance' is not aimed at analysing nuances of policies.

Note 2: Page numbers specified in the Table of Insights are as per the PDF pagination opened in a Google Chrome browser. It is possible to encounter differences in the actual page number for some of the documents when opened in a different web browser/application.

 Geography	 Data			 Policy	Aspects 			
					Healthcare Information Storage	Integrated Healthcare Management Information System		Tech-Based Digital Supply Chain System
	% of ABHAs created compared to Population of the state (Ayushman Bharat Digital Mission Dashboard) (as of 20 th October, 2023)	% of Verified Health Facilities Compared to NHRR Data under ABDM (Ayushman Bharat Digital Mission Dashboard) (as of 20 th October, 2023)	Total Consultations held through eSanjeevani (HWC & OPD) (Centre for Development of Advanced Computing) (as of Feb, 2023)	Reference Document	Maintenance of Electronic Health Records	Automation of Hospital Processes through HMIS	Integration of Human Resource Management Information System (HRMIS) within Health Sector	Integration of information system for eProcurement of medicines
Centre				Electronic Health Record (EHR) Standards for India	Yes			
				Health Management Information System		Yes		
				e-Aushadhi, Central Medical Services Society				Yes
Andhra Pradesh	80.2%	31.0%	59.85%	Electronic Health Records: Andhra Pradesh	Yes			

 Geography	 Data			 Policy	Aspects 			
	% of ABHAs created compared to Population of the state (Ayushman Bharat Digital Mission Dashboard) (as of 20 th October, 2023)	% of Verified Health Facilities Compared to NHRR Data under ABDM (Ayushman Bharat Digital Mission Dashboard) (as of 20 th October, 2023)	Total Consultations held through eSanjeevani (HWC & OPD) (Centre for Development of Advanced Computing) (as of Feb, 2023)	Reference Document	Healthcare Information Storage	Integrated Healthcare Management Information System		Tech-Based Digital Supply Chain System
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				Ayushman Bharat Digital Mission		Yes (pg. 8)		
				NHM - HRIS			Yes	
Assam	48.7%	27.6%	3.01%	NHMSSD-EHRMIS			Yes	
Chhattisgarh	62.8%	65.1%	3.13%	HRMIS Government of Chhattisgarh			Yes	
Delhi	36.6%	3.4%	0.21%	HIMS Project, Government of National Capital Territory of Delhi		Yes		
Goa	29.2%	18.2%	0.02%	e-Sushrut, Government of Goa		Yes		

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Gujarat	51.4%	29.6%	4.23%	Techno+, Government of Gujarat	Yes			
				Gujarat HMIS, Gujarat Government		Yes		
				eHRMS Gujarat			Yes	
				Gujarat Medical Services Corporation Limited, Government of Gujarat				Yes
Haryana	30.7%	7.4%	1.20%	e-Upchaar, Government of Haryana		Yes		
				HRMS Government of Haryana			Yes	

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Himachal Pradesh	49.0%	54.5%	3.85%	HMIS, Department of Digital Technologies and Governance		Yes		
				HRMS Government of Himachal Pradesh			Yes	
Jharkhand	18.1%	13.4%	1.65%	HIMS, Rajendra Institute of Medical Sciences, Government of Jharkhand		Yes		
Karnataka	37.7%	95.4%	16.79%	eHospital, Kodagu Institute of Medical Sciences, Government of Karnataka		Yes		
Kerala	40.1%	12.4%	1.84%	eHealth Kerala, Government of Kerala	Yes			
				HMIS, Government of Kerala		Yes		

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				Kerala Medical Services Corporation Limited, Government of Kerala				Yes
Madhya Pradesh	48.3%	20.0%	4.69%	eHRMs Government of Madhya Pradesh			Yes	
Maharashtra	29.3%	9.7%	3.81%	Commissionerate of Health Services of Maharashtra	Yes			
				MSPC Dawa, Government of Maharashtra				Yes
Meghalaya	17.0%	35.8%	2.28%	Meghalaya Health Policy, 2021	Yes (pg. 6)			
Odisha	55.5%	NA	0.85%	eSushrut, government of Odisha		Yes		

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				HRMS Odisha			Yes	
Punjab	23.1%	14.7%	1.18%	HIMS, Government of Punjab		Yes		
				HRMS, Government of Punjab			Yes (pg.62)	
Rajasthan	31.1%	NA	0.85%	IHMS, Government of Rajasthan		Yes		
				Rajasthan Medical Services Corporation Limited, Government of Rajasthan				Yes
Sikkim	53.7%	47.4%	2.47%	e-Sushrut, Government of Sikkim		Yes		

 Geography	 Data			 Policy	 Aspects			
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Tamil Nadu	10.2%	NA	16.15%	Tamil Nadu Health System Reform Program, Government of Tamil Nadu	Yes			
				NHM Tamil Nadu			Yes (pg. 68)	
				Tamil Nadu Medical Services Corporation Limited, Government of Tamil Nadu				Yes
Telangana	36.1%	23.6%	12.11%	eFamily, National Health Mission	Yes (pg. 92)			
				e-HMS, Government of Telangana		Yes		
				NHM HRIS			Yes (pg. 70)	

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Tripura	13.3%	4.8%	0.40%	National Health Mission Government of Tripura		Yes		
				HRMS Tripura			Yes	
Uttar Pradesh	54.6%	13.9%	2.36%	Manav Sampada Uttar Pradesh			Yes	
West Bengal	29.9%	NA	12.49%	Drug Procurement Policy, Government of West Bengal				Yes

4. List of Acronyms

- ABDM: Ayushman Bharat Digital Mission
- ABHA: Ayushman Bharat Health Account
- AI: Artificial Intelligence
- GIS: Geographic Information Systems
- HER: Electronic Health Records
- HMIS: Health Management Information System
- HRMIS: Human Resource Management Information System
- NHRR: National Health Resource Repository
- HWC: Health and Wellness Centre
- NHM-HRIS: National Health Mission Human Resources Information System
- NHMSSD-EHRMIS: National Health Mission Swasthya Sewa Dapoon Enterprise Human Resource Management Information System
- MSPC: Maharashtra State Pharmacy Council
- OPD: Outpatient Department

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